

CARTIER DUALITY AND THE LAUMON-ROTHSTEIN FOURIER TRANSFORM

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Let A be an abelian variety over a field k of characteristic 0, and let $B \cong A^\vee$ be the dual abelian scheme over k . In this short note, we explain how the the Laumon-Rothstein Fourier equivalence

$$D(\mathcal{D}_A\text{-mod}) \cong D(\text{QCoh}(B^\natural))$$

between quasi-coherent $\mathcal{D}_A$ -modules and quasi-coherent $\mathcal{O}_{B^\natural}$ -modules is a manifestation of the duality between the de Rham space A_{dR} and the universal vector extension $B^\natural$. I thank Tony Pantev for explaining several key parts of this to me.

1.1. Preliminaries. We will define the category AlgStk to be the category of fppf sheaves of k -algebras taking values in groupoids,

$$\text{AlgStk} := \text{Shv}_{\text{fppf}}(\text{Alg}_k, \text{Grpd})$$

The category of commutative k -group stacks is defined to be the subcategory $\text{CGrp} \subseteq \text{AlgStk}$ consisting of commutative group objects. We can define the following duality functor on commutative k -group stacks; let $\mathfrak{D} : \text{CGrp}^{op} \rightarrow \text{CGrp}$ be the functor

$$\mathfrak{D}(\mathcal{G}) := \underline{\text{Map}}_{\text{CGrp}}(\mathcal{G}, B\mathbb{G}_m)$$

Given a commutative k -group stack $\mathcal{G}$, we call $\mathfrak{D}(\mathcal{G})$ the *Cartier dual* to $\mathcal{G}$.

Example 1.1. For an abelian variety A , $\mathfrak{D}(A) \cong A^\vee$, the dual abelian variety. If $\mathbb{Z}$ is the constant group scheme, then $\mathfrak{D}(\mathbb{Z}) \cong B\mathbb{G}_m$ and $\mathfrak{D}(B\mathbb{G}_m) \cong \mathbb{Z}$. This is a 1-shifted or stacky analog of classical Cartier duality for group schemes.

The Cartier duality functor induces a canonical Fourier-Mukai transform. The universal evaluation morphism $\text{ev} : \mathcal{G} \times \mathfrak{D}(\mathcal{G}) \rightarrow B\mathbb{G}_m$ is equivalent to a $\mathbb{G}_m$ -torsor, and hence a line bundle $\mathcal{P}$ on $\mathcal{G} \times \mathfrak{D}(\mathcal{G})$, the *Poincaré sheaf*. This induces an integral transform

$$\Phi : D_{qc}(\mathcal{G}) \rightarrow D_{qc}(\mathfrak{D}(\mathcal{G}))$$

defined by $\mathcal{F} \mapsto R\pi_{2*}(\mathcal{P} \otimes^L \pi_1^* \mathcal{F})$ where the maps π_i are the projections

$$\begin{array}{ccc} & \mathcal{G} \times \mathfrak{D}(\mathcal{G}) & \\ \pi_1 \swarrow & & \searrow \pi_2 \\ \mathcal{G} & & \mathfrak{D}(\mathcal{G}) \end{array}$$

Under reasonable assumptions, this is an equivalence of categories. To use the terminology of [BB06], we say that $\mathcal{G}$ is *very nice* if smooth locally, $\mathcal{G}$ is isomorphic to (i) a finite product of abelian varieties over k , (ii) finitely generated abelian groups, or (iii) and copies of $B\mathbb{G}_m$. The authors prove:

Theorem 1.2 (2.7 of [BB06]). *If $\mathcal{G}$ and $\mathfrak{D}(\mathcal{G})$ are very nice, then Φ is an equivalence.*

See also [Ari04], A.1.

1.2. The Laumon-Rothstein Fourier transform. Let A be an abelian variety over k and $B = \mathfrak{D}(A)$ be the dual abelian variety. Recall that the *universal vector extension* is a smooth, commutative k -group scheme $B^\natural$ which is an extension

$$1 \rightarrow \mathbb{V}(\omega_A) \rightarrow B^\natural \rightarrow B \rightarrow 1 \quad (*)$$

where $\mathbb{V}(\omega_A)$ is the vector group of invariant differentials $\omega_A = H^0(A, \Omega_A^1)$. It's an observation due to Grothendieck that we may characterize $B^\natural$ in terms of *crystals* on A . In particular, $B^\natural$ represents the fppf sheaf $\text{Sch}_k \rightarrow \text{Set}$ given by

$$T \mapsto \{\text{isomorphism classes of degree 0 invertible crystals on } A \times T\},$$

[MM74]. Equivalently, $B^\natural \simeq \text{Pic}^\natural(A)$ is the moduli of degree-0 line bundles on A equipped with an integrable connection.

On the other hand, the de Rham space of A is the fppf sheaf with the property

$$\text{QCoh}(A_{\text{dR}}) \cong \mathcal{D}_A\text{-mod}$$

In particular, this equivalence restricts to an isomorphism $\text{Pic}^0(A_{\text{dR}}) \cong \text{Pic}^\natural(A)$. Consequently, $B^\natural$ is Cartier dual to A_{dR} . That is, there is a natural isomorphism

$$B^\natural \cong \mathfrak{D}(A_{\text{dR}})$$

Consequently, the abstract Fourier-Mukai equivalence for dual commutative group stacks recovers Laumon-Rothstein Fourier equivalence

$$\text{D}_{qc}(A_{\text{dR}}) \cong \text{D}_{qc}(B^\natural)$$

There's an even more refined statement we can make: A_{dR} fits into an exact sequence

$$1 \rightarrow \hat{A} \rightarrow A \rightarrow A_{\text{dR}} \rightarrow 1 \quad (\dagger)$$

where $\hat{A}$ denotes the formal completion of A along the identity section. Applying the duality functor $\mathfrak{D}$ to $(\dagger)$, we get an exact sequence

$$\rightarrow \underline{\text{Map}}_{\text{CGrp}}(\hat{A}, \mathbb{G}_m) \rightarrow \mathfrak{D}(A_{\text{dR}}) \rightarrow \mathfrak{D}(A) \rightarrow \mathfrak{D}(\hat{A}) \rightarrow$$

analogous to the long exact sequence in cohomology with coefficients in $\mathbb{G}_m$.

Lemma 1.3. *We have a natural isomorphism*

$$\underline{\text{Map}}_{\text{CGrp}}(\hat{A}, \mathbb{G}_m) \cong \mathbb{V}(\omega_A)$$

Proof. Since k has characteristic 0,

$$\text{Hom}_{\text{grp}}(\hat{A}, \mathbb{G}_m) \cong \text{Hom}_k(\text{Lie}(\hat{A}), \text{Lie}(\mathbb{G}_m)) \cong \text{Hom}_k(\text{Lie}(A), k)$$

and since $e^* \Omega_{A/k}^1 \cong \omega_A$ (see 2.2ii), it follows that $\text{Lie}(A)^\vee = \omega_A$. $\square$

Consequently, exact sequences $(*)$ and $(\dagger)$ are dual to each other up to a shift!

Remark 1.4. For any commutative k -group stack $\mathcal{G}$,

$$\mathrm{Lie}(\mathfrak{D}(\mathcal{G})) = H^1(\mathcal{G}, \mathcal{O}_{\mathcal{G}})$$

by a standard result of deformation theory. Thus if we apply the Lie algebra functor to the exact sequence $(*)$, we obtain

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathrm{Lie}(\mathbb{V}(\omega_A)) & \longrightarrow & \mathrm{Lie}(B^{\natural}) & \longrightarrow & \mathrm{Lie}(B) \longrightarrow 0 \\ & & \parallel & & \parallel & & \parallel \\ & & H^0(A, \Omega_{A/k}^1) & \longrightarrow & H^1(A_{\mathrm{dR}}, \mathcal{O}_{A_{\mathrm{dR}}}) & \longrightarrow & H^1(A, \mathcal{O}_A) \end{array}$$

which identifies with the Hodge filtration on $H_{\mathrm{dR}}^1(A/k)$.

Remark 1.5. After writing this short exposé, I became aware that this entire story (and much more) is already contained in the thesis of Gabriel Riberio [Rib24], who provided some very insightful comments. In particular, remark 2.3.24 provides a proof of the duality between the sequences $(*)$ and $(\dagger)$.

REFERENCES

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